

Estimating health gains from achieving optimal blood pressure control in Nepal: A rapid review and life-table analysis

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ABSTRACT

BACKGROUND Low- and middle-income countries are facing a burgeoning burden of hypertension, yet treatment and control remain critically low reflecting cascading failures across awareness, treatment access, medicine affordability, and health system capacity to provide. Simulation modelling for public health interventions is a growing field, providing crucial evidence for designing and implementing public health programs. In this study, we map existing simulation models and use lifetable modelling to provide insights into potential health gains from achieving optimal blood pressure (BP) control in Nepal, a country with a growing hypertension burden and limited primary care resources and capacity.

METHODS We searched PubMed (via Medline) with keywords combining modelling and blood pressure control, limiting to those estimating long-term cardiovascular outcomes (e.g. cardiovascular events, health adjusted life years, mortality). Further, using Nepal as a case study, we built a well-established proportional multistate lifetable model methodology– a population-level Markov state-transition model to simulate potential gains in health-adjusted life years (HALYs) from achieving SBP<140mmHg (Scenario 1), <130 (scenario 2), <120 (scenario 3) over different time horizons. Results are presented as HALYs gained with 0 and 3% discounting.

RESULTS Thirty-nine studies were included in the rapid review, with Markov state-transition models (n=12, 30.8%), microsimulation (n=11, 28.2%) and hybrid or discrete-event simulation models (n=3, 7.7%) being the most common simulation approach. Most studies focused on cardiovascular outcomes, quality adjusted life years, and cost-effectiveness. Only a few incorporated socioeconomic inequalities or were conducted in low- and middle-income settings (n=3, 7.7%). Building on this evidence base, we simulated health gains from achieving optimal BP control in low-middle-income country, Nepal. Our simulation showed that under the scenario of achieving SBP <140 mmHg among individuals with diagnosed hypertension, the model projected 1.4 (0.9 to 2.0) million HALYs gained over 20 years and 27.8 (19.0 to 39.9) million HALYs over the lifetime (undiscounted). Per-capita gains were 51.0 (35.3 to 72.5)

HALYs per 1000 population over 20 years and 1005.8 (687.9 to 1437.0) per 1000 over the lifetime. Health gains were higher among males than females (61.1 vs 41.8 HALYs per 1000 over 20 years). Sensitivity analyses indicated larger benefits with more intensive control: achieving SBP <130 mmHg generated 1.9 million and 38.5 million HALYs over 20 years and the lifetime, respectively (undiscounted). Furthermore, achieving SBP <120 mmHg generated 2.4 million HALYs over 20 years and 50.1 million over the lifetime (undiscounted).

CONCLUSION We found that achieving optimal BP control could generate substantial population health gains with more stringent targets delivering greater health benefits, providing a quantitative basis for prioritising the hypertension program within Nepal's non-communicable disease (NCD) agenda. As countries scale up the World Health Organization (WHO) HEARTS package and work toward the global BP targets (i.e., 80% diagnosed, 80% treated, 80% controlled target by 2030), country-specific models of this kind are essential to translating global ambitions into locally grounded policy.

KEYWORDS: blood pressure, policy, intervention, Nepal, hypertension, program

OSF Registration: <https://osf.io/4wyqp>

1 INTRODUCTION

Globally, 1.4 billion people are living with hypertension, of which only one in five have their blood pressure (BP) under control (1-3). Despite decades of advances in antihypertensive therapies and public health interventions, current trends remain concerning, characterised by higher prevalence and lower treatment and control rates (1). The burden is disproportionately borne by low- and middle-income countries, where health system capacity, access to medicines, and awareness of hypertension remain significant barriers to effective management (2, 4).

Achieving the World Hypertension League's 80-80-80 target by 2030 requires that 80% of people with hypertension are diagnosed, 80% of those diagnosed are treated, and 80% of those treated have controlled BP (1). This requires fourfold improvements from the current global rate (19.7%) of BP control, underscoring the urgent need for scalable, evidence-based strategies across diverse health systems. Understanding the most efficient and effective healthcare and policy options is therefore critical to guiding future BP control programs. Designing effective BP control programs requires understanding what the most efficient and appropriate health care policy options are. Simulation modelling methods has a crucial role in population intervention research for chronic diseases, as these provide a framework for understanding the dynamics of disease progression and the impact of various interventions (5, 6). However, there is limited information on simulation models currently used for estimating population health gains from interventions targeting BP control, representing an important gap in evidence to inform policy and practice.

Among low-middle-income countries, Nepal exemplifies as a country with an urgent need for concerted effort for hypertension management (7-9). The country is witnessing a rapid epidemiological transition driven by urbanisation, dietary changes

and population aging (7, 10). The prevalence of hypertension in Nepal is estimated at 25-30% among adults, which is responsible for an estimated 24% of all cardiovascular deaths (that is, over 46,000 deaths annually from stroke, ischaemic heart disease, and hypertensive heart disease combined) (11). Yet BP control rates remain critically low. Fewer than one in ten people with hypertension have their BP controlled reflecting cascading failures across awareness, treatment access, medicine affordability, and health system capacity (7-9). Simulation models can offer insights into the potential population health gains of various health and policy options that have not been evaluated in the real world due to resource limitations or ethical concerns (12). In this study, we explore the existing simulation models and use lifetable modelling to show insights on potential population health gains from achieving optimal BP control in Nepal under the standard threshold of SBP <140 mmHg, the more intensive target of SBP <130 mmHg and <120 mmHg, compared with the current business-as-usual scenario.

2 METHODS

2.1 Rapid review of existing simulation models in BP context

This study adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) extension for scoping reviews (PRISMA-ScR), with the protocol registered in Open Science Framework (registration <https://osf.io/4wyqp>). A comprehensive global search was conducted by combining search terms from earlier reviews by Feirman et al (2015) and Singh et al (2021)(13) and expanded that in the context of BP interventions (4, 14-17).

We included peer-reviewed journal articles and grey literature published from inception to July 1, 2024. We conducted searches in PubMed (via Medline), and Embase and supplemented that with searches in Google Scholar (see

eSupplementary Table 1). Additional searches included reviewing reference lists of primary studies and systematic reviews, as well as resources from relevant organizational websites. We excluded letters, case series, systematic or scoping reviews, commentaries, and editorials lacking original research findings. The study protocol was developed by the team, and a single reviewer (SRM) screened the titles and abstracts during the initial stage and conducted review of full texts in the second stage to determine eligibility. Any disagreements at either stage were resolved through discussions with GS.

A broad-based inclusion criterion was applied for inclusion of interventions aimed at BP management (either improving treatment or control rates), either conducted through government or non-governmental stakeholders. Likewise, business-as-usual was defined as current practice or continuation of the current distribution of SBP and associated cardiovascular risks in the population, with no intervention to shift population SBP distribution. Finally, the extracted variable fields included: model type, simulation time horizon, characterization of business-as-usual and interventions. We also collected information on associated health and health economic outcomes (e.g. HALYs, quality adjusted life years, cost-effectiveness) for cardiovascular diseases (e.g., myocardial infarction), cerebrovascular diseases (e.g., stroke), sudden cardiac death, and overall mortality to provide overall context to the intervention being reviewed. Where provided, data on healthcare utilization, hospitalization, and attendance at health check-ups related to cardiovascular outcomes were captured. There were no restrictions regarding population type, condition, or age.

We descriptively summarized the collected information and tabulated them in results, documenting the availability of simulation models by model types, and similarities and differences between these models. Three reviewers (SRM, KB, BA) synthesized the

findings from the included articles using both tables and a narrative summary of main findings. The findings were organized into three main themes: 1) model types, 2) specifications and 3) disease areas. Where relevant we added modelling results to provide more context to the model being studied. We did not assess the quality of the studies as methodological quality and risk of bias have limited relevance to conceptualization and are not recommended for scoping review methodology.

2.2 Lifetable analysis

We used the well-established proportional multistate lifetable modelling methodology to estimate the lifetime health gains from achieving optimal BP control in the adult Nepalese population in 2023 (16-21), hereafter mentioned as BP-MSLT. We simulated multiple cohorts of population into the future (16), and compared several counterfactual scenarios to estimate the potential health gains associated with progressively stricter BP control. In our study, three intervention scenarios were modelled to reflect progressively more intensive SBP control targets. In the first scenario (standard control), individuals with SBP ≥ 140 mmHg were assumed to receive treatment sufficient to reduce their SBP to < 140 mmHg. In the second scenario (intensive control), individuals with SBP ≥ 130 mmHg were assumed to achieve SBP < 130 mmHg. In the third scenario (very intensive control), SBP was reduced to < 120 mmHg among individuals with SBP ≥ 120 mmHg.

2.2.1 Business-as-usual characterisation

The business-as-usual SBP distribution for each sex by age by socioeconomic status (SES) group was estimated using data from the Nepal Demographic Health Survey 2022 (*see the supplementary Table 5 for SBP distribution by age and sex*). The rate

ratios associating SBP with disease incidence were taken from the GBD 2023 (ghdx.healthdata.org).¹⁷

2.2.2 PIFs to link BP intervention and the life table model

The study uses the relative risk (RR) shift method for calculating the population impact fraction (20), whereby people stay in their 'starting state' but shift the mean SBP of their state and are then assigned the RR for the new mean SBP. In our simulation, for each 10-unit SBP category, ≥ 140 mm/Hg is 'shifted' to the RR for SBP of < 140 mm/Hg. For each sex by age group, a population impact fraction (PIF) was calculated for modelled diseases. The generic formula is available elsewhere (22). These PIFs are the percentage change in incidence rates for each SBP-related disease, by socio-demographics and time step into the future, that are inputted to the BP-MSLT. We applied a median lag time of 3 years for CVD (In normal $\pm 20\%$) (23). These PIFs serve as a bridge between SBP-related relative risks and diseases specific rates. We included all SBP-related diseases that contributed to over 90% of all DALY loss from SBP in Nepal in 2023 (24).

2.2.3 Analysis

We modelled 5 independently associated SBP-related diseases (Figure 1), simulated separately for male and female cohorts from baseline age through to death under: (1) business-as-usual (and (2) intervention (optimal BP attained and sustained)).

Transition rates between health states (healthy, cardiovascular disease event, post-event, CVD death, non-CVD death) were derived from Nepal-specific incidence, case-fatality and mortality data where available from GBD and supplemented by regional estimates. Relative risk reduction for 10-mmHg reduction in cardiovascular incidence from shifting BP distribution were obtained from GBDx. The BP-MSLT projected changes in health-adjusted life-years (HALYs) for each scenario. Uncertainty was

addressed by Monte Carlo simulation (2,000 iterations), drawing from distributions around BP-associated relative risks, and baseline morbidity and mortality disease rates. Results are presented as incremental HALYs gained per 2,000 adults under the intervention scenario relative to business-as-usual. Each model run was run 2000 times, propagating uncertainty about all input parameters (see Table 1 and 2). Models were run with both 0% and 3% annual discount rates. Analyses were conducted using Python version 3.9.5 (Delaware, USA).

2.2.4 Sensitivity analysis

First, we conducted two sensitivity analyses by changing the SBP control levels to more intense targets of 120 and 130 mmHg. Second, given the lack of data to estimate the rate ratio difference in morbidity and mortality rates by SES in Nepal, we did not stratify results by socio-economic quintiles.

3 RESULTS

3.1 Evidence from prior studies

A total of 39 studies examining simulation models of BP interventions were identified (see Figure 1). The most frequently reported outcomes were economic outcomes, including incremental cost-effectiveness ratios (ICERs), cost per QALY, or net cost savings, in 25 studies (64.1%), and cardiovascular events such as myocardial infarction or stroke in 16 studies (41.0%). Health outcome metrics included quality-adjusted life years (QALYs) in 15 studies (38.5%), mortality outcomes in 10 studies (25.6%), and disability-adjusted life years (DALYs) in six studies (15.4%). Life-years gained or saved were reported in five studies (12.8%). Net cost savings where averted expenditure was explicitly quantified were reported in eight studies (20.5%). Blood pressure outcomes themselves were reported in four studies (10.3%), while dementia outcomes were reported in two studies (5.1%).

Panel 1. Summary of models used in prior studies

What types of models were used?

All 39 studies employed a classifiable modelling framework. Markov state-transition models were the most common approach (n=12, 30.8%), followed by microsimulation (n=11, 28.2%) and hybrid or discrete-event simulation models (n=3, 7.7%). Population-scale simulation models, including adaptations of the CVD Policy Model, were used in two studies (5.1%). A number of additional modelling approaches were each reported in single studies, including spreadsheet-based deterministic models, quasi-experimental combined with cost-effectiveness analysis, system dynamics models, comparative risk assessment, causal inference models, cost-avoidance models, decision-analytic models, and longitudinal population models.

How were models parameterised?

The most commonly included input parameters were blood pressure levels (n=26, 66.7%), age (n=17, 43.6%), and other cardiometabolic risk factors such as cholesterol (n=10, 25.6%), body height and weight or BMI (n=9, 23.1%), diabetes (n=6, 15.4%), smoking (n=5, 12.8%), and treatment adherence or access (n=6, 15.4%). Fewer studies incorporated contextual or social determinants: income or education (n=4, 10.3%), ethnicity (n=3, 7.7%), and urban–rural location (n=2, 5.1%). Majority of the studies included costs (n=22, 56.4%) as input parameters.

How were models specified?

Most models used annual time steps (n=29, 74.4%), with three studies using six-month cycles and five using no time step (cross-sectional or not applicable). Time horizons ranged widely: 12 studies (30.8%) used a lifetime horizon, 11 studies (28.2%) used a 10-year horizon, and the remainder used horizons ranging from one to 30 years. Population heterogeneity was incorporated through stratification by sex (n=18, 46.2%) and age (n=16, 41.0%) in about half of studies. Other stratification dimensions included ethnicity (n=10, 25.6%), income or education (n=10, 25.6%), baseline blood pressure level (n=8, 20.5%), and other cardiovascular risk factors (n=8, 20.5%). Urban–rural stratification was not reported in any study.

3.2 Health gains from achieving optimal BP control in Nepal: Case study

Achieving optimal BP control in Nepal generated substantial population health gains.

Under the scenario in which all individuals achieved SBP <140 mmHg, the model projected 1.4 million HALYs gained over the first 20 years (95% UI: 0.9 to 2.0 million) and 27.8 million HALYs over the lifetime (95% UI: 19.0 to 39.9 million) (undiscounted).

When applying a 3% annual discount rate, estimated gains were 1.0 million HALYs over 20 years and 6.8 million HALYs over the lifetime.

In per-capita terms, this corresponds to 51.0 HALYs gained per 1000 population over 20 years (95% UI: 35.3 to 72.5) and 1005.8 HALYs per 1000 population over the lifetime (95% UI: 687.9 to 1437.0).

<Insert Table 1>

3.3 Sex differences in HALY gains in Nepal

Health gains differed by sex, with larger benefits observed among males. Over the first 20 years, males gained 61.1 HALYs per 1000 population (95% UI: 42.0 to 85.8)

compared with 41.8 HALYs per 1000 among females (95% UI: 28.5 to 59.4). Over the lifetime horizon, gains were 1142.7 HALYs per 1000 males (95% UI: 777.4 to 1617.5) and 872.1 HALYs per 1000 females (95% UI: 589.5 to 1237.9).

3.4 Sensitivity analysis: more intensive BP targets

More intensive BP targets produced larger health gains. In the scenario achieving SBP <130 mmHg, the model estimated 1.9 million HALYs gained over 20 years (95% UI: 1.3 to 2.7 million) and 38.5 (26.6 to 54.7) million HALYs over the lifetime (undiscounted); discounted gains were 1.3 (0.9 to 1.8) million HALYs over 20 years and 9.4 million HALYs over the lifetime. Per-capita gains were 69.5 HALYs per 1000 population over 20 years (95% UI: 50.0 to 97.7) and 1386.1 HALYs per 1000 population over the lifetime (95% UI: 975.5 to 1970.3). Under the most intensive BP control scenario of SBP <120 mmHg, gains reached 2.4 million HALYs over 20 years (95% UI: 1.8 to 3.4 million) and 50.1 million over the lifetime (undiscounted); discounted gains were 1.6 million and 12.1 million HALYs, respectively. Per-capita gains were 89.4 HALYs per 1000 population over 20 years (95% UI: 65.4 to 120.7) and 1813.2 per 1000 over the lifetime (95% UI: 1277.8 to 2527.3). In all sensitivity scenarios, gains were consistently larger among males than females (eSupplementary Tables 8).

4 DISCUSSION

4.1 Summary of main results

This study provides two key insights. First, our review identified substantial diversity in simulation models used to evaluate BP interventions, with Markov models being the most common. Most studies focused on cardiovascular outcomes, quality adjusted life years, and cost-effectiveness. Only a few incorporated socioeconomic inequalities or were conducted in low- and middle-income settings, a gap that directly reflects

inequities in the global evidence base for hypertension policy. Second, using Nepal as a case study, our life-table model estimated that achieving optimal BP control could generate large population health gains, with benefits increasing substantially under more intensive SBP targets. Notably, modelled health gains were consistently higher among men than women. This likely reflects a combination of factors: men in Nepal tend to have higher burden of hypertension and downstream CVD events, but simultaneously face lower rates of hypertension awareness (men vs women = 48% vs 52.3%), treatment (28.0% vs 35.2%) and control rates (16.4% vs 19.5%), meaning the counterfactual benefit of achieving control is larger among men(10). Together, these factors result in a larger absolute reduction in cardiovascular risk when control is achieved, driving higher modelled health gains among men despite their shorter life expectancy at birth compared to women (men vs women = 69 vs 72 years).

4.2 Comparison of study results

Our study used a novel counter-factual model that simulates the health gains from achieving optimal blood pressure control in Nepal, which could ultimately help in tracking Nepal's progress towards World Hypertension League's 80-80-80 target by 2030, which requires that 80% of people with hypertension are diagnosed, 80% of those diagnosed are treated, and 80% of those treated have controlled BP (1).

Globally, achieving this target requires substantial improvement from the current BP control rate of approximately 19.7%, with only 9.3% of people with hypertension achieving control in low-income countries compared to 28.3% in high-income countries (2). A global modelling estimated that under realistic implementation conditions, achieving the 80-80-80 target could reduce all-cause mortality by 4 to 7%, averting 76–130 million deaths over 2022–2050(25). Our Nepal-specific findings modelled health gains under three blood pressure control targets (<140, <130, <120

mmHg), providing country-level HALY estimates grounded in nationally representative survey data, and by modelling outcomes across 20-year, and lifetime horizons. This extended horizon is methodologically important: most cost-effectiveness analyses use shorter time frames that may systematically undervalue preventive interventions whose benefits accrue over decades.

Achieving optimal blood pressure control would require drastic improvement in hypertension care cascade; improving health system capacity to provide hypertension services and removing barriers that limit access to this service, improves health literacy and medication adherence. Analysis of the 2022 NDHS found that among hypertensive adults, only 50.2% were aware of their diagnosis, 31.7% were receiving treatment, and 18.0% had controlled BP (26). These figures represent a formidable gap from the 80-80-80 target and reflect well-documented structural barriers, including limited primary care capacity, medicine availability, and out-of-pocket costs. Studies from low- and middle-income countries (LMICs) indicate that antihypertensive medications cost more than 10% of a typical government worker's monthly salary, with only 7–26% availability of basic cardiovascular medicines in the public sector (27). The closest comparable LMIC modelling evidence comes from Indonesia, where estimates of gains in adult life expectancy from improving SBP control suggested that realistic hypertension interventions would likely produce more modest gains than theoretical optimal scenarios, underscoring the importance of comparing strategies and identifying cost-effective implementation pathways (28). A recent simulation across 44 LMICs found that greater health benefits accrued to the bottom wealth quintiles in countries with larger baseline disparities in hypertension management, and that LMICs would see the greatest absolute benefits from improvements in the treatment cascade (29, 30). While health gains across SES quintiles were not

evaluated in this study, we would anticipate greater gains among lower SES groups, where hypertension, though potentially less prevalent, is more likely to be undetected, untreated and poorly controlled – resulting in disproportionately higher cardiovascular mortality and morbidity rates and therefore greater absolute benefit from improved care. This research question needs further investigation. The relationship between socioeconomic position and hypertension-attributable burden in Nepal remains poorly characterised, and future research incorporating SES-stratified rate estimates would be valuable to quantify these inequities and inform more targeted intervention strategies.

Despite the majority of hypertension-related deaths concentrated in low-income countries, there are only a few studies estimating the health gains of broader public health interventions in this region. This could be explained in terms of a sheer lack of simulation models, which supports our argument that technical and computing capacity remain a hindering factor for wider use of simulation models (31). This analysis for Nepal is among the first such studies conducted in a low-income country, and a clearly important step in advancing this methodology in low-income settings.

4.3 Policy implications

These findings have direct relevance for Nepal's national NCD strategy and health financing priorities. The substantial modelled HALY gains, under three scenarios, highlight the sheer 'opportunity cost' of inaction and can inform the Multisectoral Action Plan for the Prevention and Control of NCDs (2026-2030), a landmark policy initiative to curb the growing burden of NCDs in Nepal (32). Addressing the population-level burden of elevated blood pressure in Nepal will require a dual strategy that extends beyond the screening and treatment of individuals with clinically defined hypertension targets. While improving the hypertension care cascade remains essential, the

majority of BP-attributable cardiovascular events arise from the large proportion of the population with modestly elevated but sub-threshold blood pressure, a phenomenon well described by Rose's prevention paradox. (33) Therefore, population-wide, non-pharmacological strategies targeting the upstream determinants of blood pressure, including dietary sodium reduction (34), physical activity promotion (35), and policies to reduce harmful alcohol use (36), are a necessary complement to high-risk clinical approaches.

The WHO HEARTS technical package offers a promising implementation framework for scaling hypertension control in LMICs (37). By the end of 2024, 38 LMICs were participating in the Global Hearts programme, with more than 34 million patients treated and improved hypertension control outcomes registered over time (38, 39). Population health modelling of the kind presented here provides the evidence base that policymakers require to make the investment case for scaling such programmes, quantifying the health gains achievable under different intervention scenarios and identifying where along the care cascade efforts should be prioritised. Here, we identify opportunities to invest in primary and community care for hypertension as a high-impact, system-strengthening strategy to advance national NCD targets and reduce premature cardiovascular mortality, while underscoring the importance of country-specific simulation models using locally representative data to guide national scale-up of the HEARTS package.

Although we did not stratify health gains by socioeconomic status in the current analysis, socioeconomic inequalities in hypertension awareness, treatment, and control are pronounced in Nepal: those in the wealthiest category are 5.1 times more

likely to receive treatment than those in the poorest category(10), Therefore, this represents an important priority for future work.

4.4 Limitations

Our study has some limitations, which are typical of any life table studies (16, 17).

First, our rapid review is limited by the breadth of literature searched and the exclusion of non-English language studies, which may have resulted in the omission of relevant Nepal-specific evidence, particularly grey literature and locally published data on hypertension burden and care cascade performance. Second, BP inputs were drawn from clinically measured data in the NDHS using a standardised three-reading protocol, measurements were taken on a single occasion during a household survey visit rather than across repeated clinical encounters as recommended for formal hypertension diagnosis; this is unlikely to substantially bias population-level model inputs. Second, the simulation model does not incorporate new births or migration. Excluding births will have little impact on estimated HALY gains over a 20-year horizon, given that BP-related disease burden is concentrated in middle and older age groups; however, ongoing population growth through migration may mean health gains are underestimated. Third, the BAU scenario assumes the BP distribution observed at baseline persists throughout the projection period. Given Nepal's rapid epidemiological transition and rising hypertension prevalence, HALY gains from achieving optimal BP control may be underestimated if the comparator population becomes more hypertensive over time. Fourth, the model does not capture mental health gains associated with BP reduction, as these are not included in GBD-derived outcome estimates. Fifth, we did not estimate health gains by socioeconomic status; given Nepal's pronounced socioeconomic inequalities in both hypertension prevalence and access to treatment, future work should examine the distribution of health gains across SES strata to inform equity-focused policy.

5 CONCLUSIONS

We demonstrated that achieving optimal BP control in Nepal could generate substantial population health gains, with benefits scaling under more ambitious intervention targets. These findings provide a quantitative basis for prioritising hypertension control within Nepal's national NCD agenda and health financing decisions. As countries scale up implementation of the WHO HEARTS technical package and work toward achieving 80-80-80 target, country-specific population health models of this kind will be essential to translating global ambitions into locally grounded, equity-informed policy. Future work should extend this modelling framework to estimate health gains disaggregated by socioeconomic status, enabling policymakers to quantify the equity dividend of reaching the poorest and most underserved populations first.

Conflict of Interests

AES received speaker honoraria and/or consultancy fees from Servier, Abbott, Sanofi, AstraZeneca, Medtronic, Omron, Aktia and Sky Labs. Other authors declare no conflict of interests.

Data Availability

Data used in this study is available upon reasonable request.

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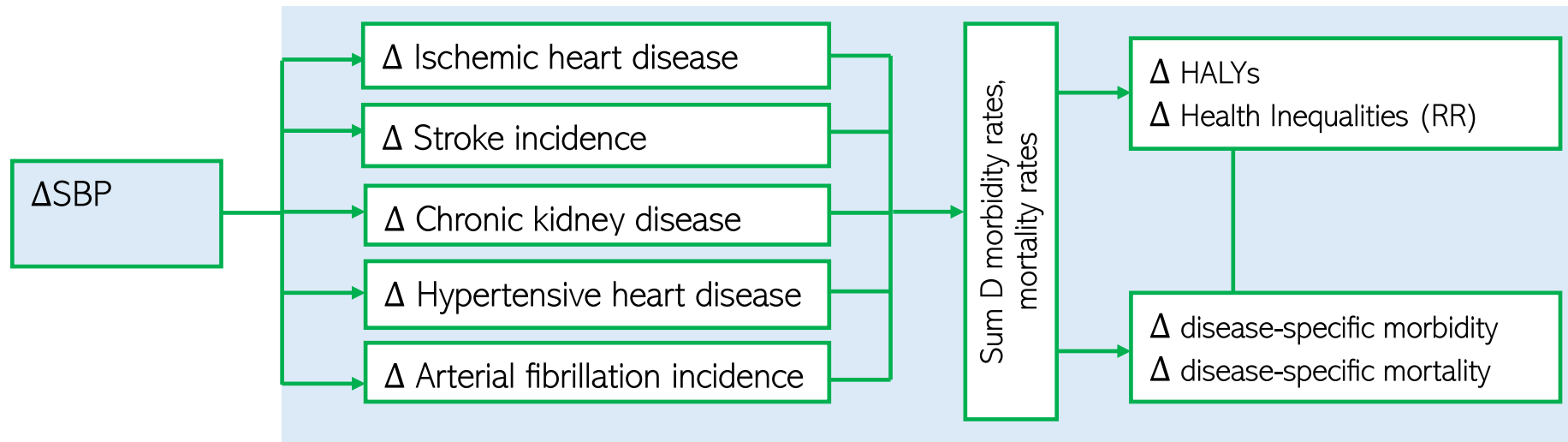
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Figure 2. Conceptual diagram for the relationship between blood pressure and health outcomes



INTERVENTION SCENARIOS

1. Clinical scenario [SBP <140 mmHg]: Standard control scenario where individuals with SBP ≥ 140 mmHg are assumed to receive treatment to reduce SBP below 140 mmHg.
2. Intensive control scenario [SBP <130 mmHg]: individuals with SBP ≥ 130 mmHg are assumed to receive treatment to reduce SBP below 130 mmHg.
3. Very intensive control scenario [SBP <120 mmHg]: where individuals with SBP ≥ 120 mmHg are assumed to receive treatment to reduce SBP below 120 mmHg.

Table 1. Future HALYs gained from achieving 100% hypertension control (SBP <140 mmHg) among individuals with diagnosed hypertension compared to BAU in Nepal.

	Undiscounted (95% UI) ^a		Discounted (95% UI) ^a at 3% per annum	
	2023-2043	Lifetime	2023-2043	Lifetime
Health adjusted life years				
Total HALYs (millions)	1.4(0.9 to 2.0)	27.8 (19.0 to 39.9)	1.0 (0.6 to 1.3)	6.8 (4.5 to 9.6)
Total HALYs per 1000 people in 2023	51.0 (35.3 to 72.5)	1005.8 (687.9 to 1437.0)	33.7 (23.2 to 47.8)	245.0 (169.8 to 346.5)
By sex				
Females, per 1000 people in 2023	41.8 (28.5 to 59.4)	872.1 (589.5 to 1237.9)	27.5 (18.8 to 39.3)	204.8 (139.3 to 292.0)
Males, per 1000 people in 2023	61.1 (42.0 to 85.8)	1142.7 (777.4 to 1617.5)	40.3 (27.5 to 56.6)	286.2 (197.5 to 401.8)

a. point estimates with 95% UI in brackets.

Disease	20-YEAR HORIZON			LIFETIME HORIZON		
	All	Female	Male	All	Female	Male
Stroke	431k (223k-727k)	197k (101k-324k)	235k (122k-397k)	3.06M (1.60M-4.98M)	1.41M (738k-2.29M)	1.65M (864k-2.69M)
Ischaemic heart disease	277k (158k-491k)	104k (58k-184k)	173k (100k-307k)	2.33M (1.35M-4.04M)	881k (521k-1.51M)	1.45M (832k-2.56M)
Hypertensive heart disease	164k (47k-272k)	62k (15k-112k)	101k (31k-153k)	1.06M (271k-1.66M)	453k (97k-745k)	599k (173k-911k)
Chronic kidney disease	17k (9k-28k)	8k (4k-13k)	9k (5k-15k)	106k (52k-176k)	50k (25k-83k)	56k (28k-97k)
Atrial fibrillation	26k (14k-45k)	12k (7k-21k)	13k (8k-24k)	151k (87k-249k)	75k (42k-127k)	76k (45k-126k)

Figure 3. Estimated health gains (HALYs) and 95% UI at 3% annual discount rate from achieving optimal blood pressure control (<140 mmHg) in Nepal, by disease and time horizon (2023–2043 and lifetime). HALYs = health-adjusted life years.

SUPPLEMENTARY MATERIAL

eSupplementary Figure 1. Flow diagram

eSupplementary Table 1. Search strategy

eSupplementary Table 2. Descriptive characteristics of reviewed studies: input parameters, heterogeneity, outcomes reported and modelling methods in reviewed studies

eSupplementary Table 3. Simulation models with potential applications for assessing health gains from BP control interventions

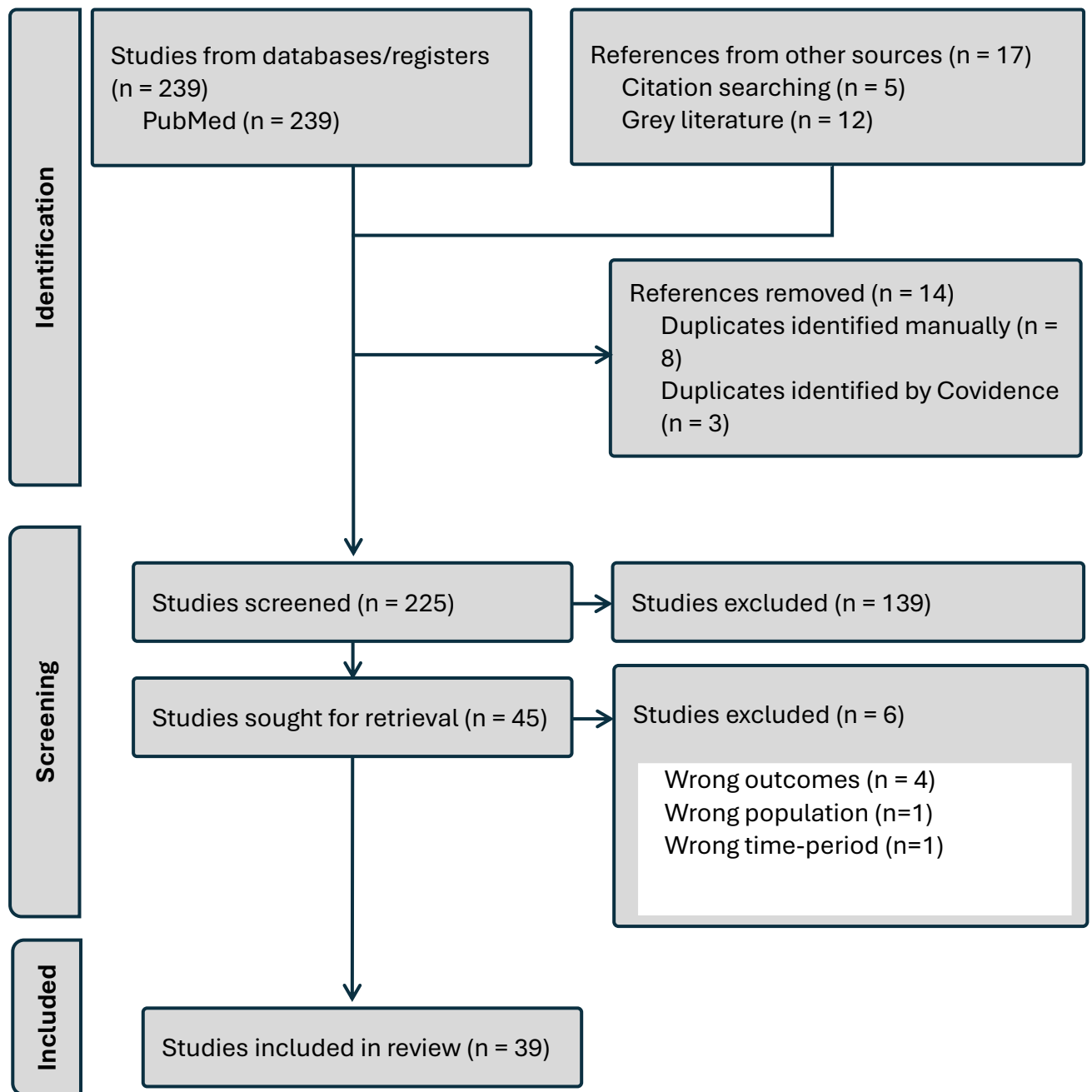
eSupplementary Table 4. Study and Model Characteristics

eSupplementary Table 5. Model Parameters and Results

eSupplementary Table 6. SBP distribution by 5-year age groups and sex based on NDHS 2022.

eSupplementary Table 7. Business-as-usual input parameter and epidemiological parameters and their sources for the base year (i.e. 2025) and projections.

eSupplementary Table 8. Future HALYs gained from achieving 100% hypertension control (SBP <120/120 mmHg and <130 mmHg) among individuals with diagnosed hypertension compared to BAU in Nepal.



eSupplementary Figure 1. Flow diagram for the rapid review

eSupplementary Table 1. Search strategy

Concept 1: modelling	Concept 2: blood pressure	Restrictions
<i>PubMed</i> 'models, theoretical'[majr:noexp] or 'models, statistical'[majr:noexp] or 'models, economic'[majr] or 'computer simulation'[majr:noexp] or 'monte carlo method'[mesh] or 'decision support techniques'[majr:noexp] or 'decision trees'[mesh] or 'systems theory'[mesh] or 'markov chains'[mesh] or 'system dynamics'[tiab] or 'agent-based model'[tiab] or 'agent-based models'[tiab] or 'agent-based modeling'[tiab] or 'agent-based modelling'[tiab] or 'simulation model'[tiab] or 'decision analysis'[tiab] or 'decision framework'[tiab] or 'markov'[tiab] or 'cost-utility analysis'[tiab] or 'cost-utility analyses'[tiab] or 'cost-effectiveness analysis'[tiab] or 'cost-effectiveness analyses'[tiab] or 'cost-benefit analysis'[tiab] or 'cost-benefit analyses'[tiab] or 'forecasting'[mesh] or 'microsimulation'[tiab] or 'micro simulation'[tiab] or 'monte carlo'[tiab] or 'life year'[tiab] or 'life years'[tiab] or 'smoking-attributable deaths'[tiab] or 'smoking attributable deaths'[tiab] or 'deterministic'[tiab] or 'probabilistic'[tiab] or 'stochastic'[tiab] or 'dynamic transmission model'[tiab] or 'state-transition'[tiab] or 'state transition'[tiab] or 'discrete event'[tiab] or 'continuous event'[tiab] or 'analytic horizon'[tiab] or 'cohort simulation'[tiab] or 'second-order simulation'[tiab] or 'first-order simulation'[tiab] or 'threshold analysis'[tiab] or 'years of healthy life'[tiab] or 'decision problem'[tiab] or 'transition probabilities'[tiab] or 'discount rate'[tiab]	("Blood Pressure"[Mesh] OR "Cardiovascular Diseases"[Mesh] OR "Blood Pressure Control" OR "Hypertension Control") AND ("Intervention" OR "Program" OR "Policy") AND ("Treatment" OR "Management" OR "Therapy")	NOT 'genetic'[TIAB] NOT 'gene'[TIAB] Filter: English
<i>Embase</i> 'theoretical model'/mj OR 'statistical model'/mj OR 'computer simulation'/mj OR 'disease simulation'/mj OR 'monte carlo method'/mj OR 'decision support system'/mj OR 'decision tree'/mj OR 'systems theory'/mj OR 'economic evaluation'/exp OR 'forecasting'/exp OR 'economic model':ab,ti OR 'simulation model':ab,ti OR 'markov':ab,ti OR 'systems dynamics':ab,ti OR 'agent-based model':ab,ti OR 'agent-based models':ab,ti OR 'agent-based modeling':ab,ti OR 'agent-based modelling':ab,ti OR 'decision analysis':ab,ti OR 'decision framework':ab,ti OR 'microsimulation':ab,ti OR 'micro simulation':ab,ti OR 'life year':ab,ti OR 'life years':ab,ti OR 'smoking-attributable deaths':ab,ti OR 'smoking attributable deaths':ab,ti OR	("Blood Pressure"[Mesh] OR "Blood Pressure Control" OR "Hypertension Control") AND ("Intervention" OR "Program" OR "Policy") AND ("Treatment" OR "Management" OR "Therapy")	NOT 'gene'/exp OR 'genetic'/exp AND [english]/lim AND [humans]/lim

eSupplementary Table 2. Descriptive characteristics of reviewed studies: input parameters, heterogeneity, outcomes reported and modelling methods in reviewed studies

Variables	N (%)
Total	39 (100)
Input parameters ^a	
Costs	22 (56.4)
Blood pressure	26 (66.7)
Age	17 (43.6)
Body height and weight	10 (25.6)
Cholesterol	9 (23.1)
Diabetes	6 (15.4)
Sex	7 (17.9)
Treatment adherence or access	6 (15.4)
Smoking	5 (12.8)
Ethnicity	4 (10.3)
Income or education	3 (7.7)
Urban or rural	2 (5.1)
Stratification ^a	
Age	18 (46.2)
Sex	16 (41.0)
Ethnicity	10 (25.6)
Income or education	10 (25.6)
Blood pressure level	8 (20.5)
Other risk factors	8 (20.5)
Urban or rural	0 (0)
Outcomes reported ^a	
Cost savings	25 (64.1)
Cardiovascular events (myocardial infarction, stroke)	16 (41.0)
Quality adjusted life years	15 (38.5)
Cost-effectiveness ratio	12 (30.8)
Mortality	10 (25.6)
Blood pressure outcomes (control/reduction)	6 (15.4)
Disability adjusted life years	5 (12.8)
Dementia	4 (10.3)
Modelling methods	
Markov / state-transition model	12 (30.8)
Microsimulation	11 (28.2)

Variables	N (%)
Hybrid / discrete-event simulation	3 (7.7)
Population-scale simulation (CVD Policy Model etc.)	2 (5.1)
Spreadsheet / deterministic model	1 (2.6)
Quasi-experimental + CEA	1 (2.6)
System dynamics model	1 (2.6)
Comparative risk assessment	1 (2.6)
Causal inference model	1 (2.6)
Cost-avoidance model	1 (2.6)
Decision analytic model	1 (2.6)
Longitudinal population model	1 (2.6)
Monte Carlo sampling / imputation	1 (2.6)
Framingham-based / cross-sectional CUA	2 (5.1)
Not specified	0 (0)

^aCategories are not mutually exclusive; studies may report multiple parameters, subgroups, or outcomes. Values in red bold indicate corrected figures. Values in grey italic indicate the original (incorrect) figures. Abbreviations: CEA = cost-effectiveness analysis; CUA = cost-utility analysis; DALY = disability-adjusted life year; ICER = incremental cost-effectiveness ratio; MI = myocardial infarction; QALY = quality-adjusted life year.

eSupplementary Table 3. Simulation models with potential applications for assessing health gains from BP control interventions

Model Type	What it is typically used for	Core features	Key limitations	Application for BP modelling	Key references
Markov (State-Transition) Models N=7	Used to represent how population move between health states (e.g., healthy, disease, death) over time, particularly for chronic diseases	Defined health states; transition probabilities; cycle length; utilities; costs; discounting	Assumes that future transitions depend only on the current state ("memoryless" property); limited ability to account for complex disease histories	This can be used to model long term health outcomes of BP control.	Sonnenberg FA, Beck JR. <i>Med Decis Making</i> . 1993;13:322–338. doi:10.1177/0272989X9301300409. Siebert U et al. <i>Med Decis Making</i> . 2012;32:690–700. doi:10.1177/0272989X12458348
Microsimulation Models N=9	Model disease progression at the individual level, allowing each simulated person to have unique characteristics and risk trajectories	Individual risk factors; demographics; event history; stochastic transitions; treatment effects	Computationally demanding; requires detailed data; results can be sensitive to modelling assumptions	This can be used to model individual BP trajectories over time, accounting population heterogeneity.	Krijkamp EM et al. <i>Med Decis Making</i> . 2018;38:400–422. doi:10.1177/0272989X18754513
Discrete Event Simulation (DES) N=2	Model systems where events occur at specific points in time (e.g., clinic visits, hospital admissions, treatment delays)	Time-to-event modelling; queuing systems; resource use; patient pathways	Requires detailed time-based data; can be complex to implement	Modelling to optimise clinical workflows/service delivery and resource allocation	Karnon J et al. <i>Value Health</i> . 2012;15:821–827. doi:10.1016/j.jval.2012.04.013. Jun JB et al. <i>J Oper Res Soc</i> . 1999;50:109–123.
System Dynamics Models N=0	Explore how diseases and interventions evolve at the population level over time, especially where feedback mechanisms are important	Stock-and-flow structures; feedback loops; aggregated population parameters; scenario testing	More abstract; less suited for modelling individual variability; parameter uncertainty can influence results	This can be used to model hypertension care cascade as stock flows, accounting service and workforce constraints.	Homer JB, Hirsch GB. <i>Am J Public Health</i> . 2006;96:452–458. doi:10.2105/AJPH.2005.062059
Agent-Based Models (ABM) N=0	Simulate interactions between individuals and their environment, often used when behaviour and social	Behavioural rules; social interactions; adaptive responses; heterogeneous agents	Complex to parameterise and validate; computationally intensive; outcomes sensitive to behavioural assumptions	Can be used to model behaviour change and social policy and impact on BP control	Auchincloss AH, Diez Roux AV. <i>Am J Epidemiol</i> . 2008;168:1–8. doi:10.1093/aje/kwn118. Epstein JM, Axtell R. <i>Growing Artificial Societies</i> . MIT Press; 1996.

	networks influence outcomes				
Compartmental Models N=0	Model how populations move between defined compartments (e.g., susceptible, diseased, treated), often used in infectious disease but also applicable to chronic conditions	Compartment sizes; transition rates; mixing assumptions	Assumes homogeneous mixing; does not capture individual-level heterogeneity	Modelling hypertension care cascade, where intervention changes diagnosis, treatment and control rates. Modelling a mix of individual and system-level dynamics, suitable for complex multicomponent programs.	Kermack WO, McKendrick AG. <i>Proc R Soc Lond A</i> . 1927;115:700–721. doi:10.1098/rspa.1927.0118. Hethcote HW. <i>SIAM Rev</i> . 2000;42:599–653. doi:10.1137/S0036144500371907
Hybrid Models N=0	Combine two or more modelling approaches (e.g., DES with system dynamics, or agent-based with Markov) to capture complex systems more realistically	Integration of multiple modelling structures; cross-model data exchange	Increased structural complexity; more difficult to validate; higher computational demands		Brailsford SC et al. <i>Eur J Oper Res</i> . 2019;278:721–737. doi:10.1016/j.ejor.2018.10.025

eSupplementary Table 4. Study and Model Characteristics

Author (Year)	Country	Model Type	Scope	Time Steps	Time Horizon	Sensitivity Analysis
Almalki et al. (2020)	Saudi Arabia	Markov state-transition model	Cost-effectiveness of intensive (<120 mmHg) vs. standard (<140 mmHg) SBP treatment in high-risk hypertensive patients without diabetes	Annual	5–30 years	One-way SA: costs, probabilities, utilities PSA (Monte Carlo)
Ambavane et al. (2019)	USA	Microsimulation	Stroke incidence and mortality under transcendental meditation intervention in adults ≥25 with SBP >140 mmHg over 15 years	Annual	15 years	One-way and two-way SA
Basu et al. (2016)	China and India	Microsimulation	Compares benefit-based tailored treatment (BTT), treat-to-target (TTT), and hybrid BP strategies for CVD mortality reduction	Annual	10 years	PSA
Boersma et al. (2007)	Netherlands	Cost-effectiveness analysis	Cost-effectiveness of losartan vs. atenolol for stroke prevention (LIFE trial data)	5.5-year trial + lifetime extrapolation	Lifetime	Varied discount rates stroke costs varied bootstrapping (1,000 simulations)
Bryant et al. (2021)	USA	Hybrid discrete-event simulation	10-year costs and health gains of LABBPS pharmacist intervention in hypertensive Black men	Annual	10 years	One-way SA three adherence/duration scenarios PSA: 500 iterations
Bryant et al. (2023)	USA	Hybrid discrete-event simulation	Long-term BP outcomes, CVD events, healthcare costs, survival, and QALYs under team-based care	Annual	10 years	Varied TBC visit frequency, medication titration, TBC duration (1, 5, 10 years)
Burke et al. (2024)	USA	Microsimulation	MI, stroke, dementia, cognitive performance, and mortality under population-level BP treatment strategies	Annual	Lifetime	Varied baseline BP matched population analysis dementia risk distributions
Burke et al. (2024)	USA	Microsimulation	ASCVD, dementia, and mortality across the life course individual- and population-level BP treatment strategies	Annual	Lifetime	Varied BP intensity/timing dementia recalibration usual vs. intensive BP control QALY gains by BP quartile
Cazarim et al. (2018)	Brazil	Quasi-experimental CEA	Cost-effectiveness of pharmaceutical care vs. conventional care in Brazil's public health system	Annual	7 years	Monte Carlo (10,000 iterations) one-way SA
Chay et al. (2024)	Kenya	Markov model	Group medical visits, microfinance, and combined intervention for hypertension (BIGPIC trial data)	Annual	10 years (20-year SA)	One-way deterministic (tornado) PSA (10,000 iterations) CEACs
Choi et al. (2016)	USA	Discrete-time microsimulation Monte Carlo	Impact of expanding NSRI to all food producers on hypertension, MI, stroke, and mortality	Annual	10 years	Salt sensitivity by hypertension status/race partial producer participation consumer salt addition

Author (Year)	Country	Model Type	Scope	Time Steps	Time Horizon	Sensitivity Analysis
Fishman et al. (2013)	USA	CEA microsimulation (life expectancy)	Cost-effectiveness of usual care, home BP monitoring, and home BP monitoring + web-based pharmacist care	Annual	Lifetime	Bootstrap 95% CIs discount rates 3%, 5%, 7% varied life expectancy gains
Fonseca et al. (2023)	30 countries	Microsimulation	CVD events and costs from cuff SBP inaccuracies (under/overestimation) vs. invasive aortic SBP	Annual	1 year	Stratification by SBP category and degree of discordance imputation inflation adjustment
Galani et al. (2007)	Switzerland	Markov model CUA PSA	Lifetime cost-effectiveness of 3-year lifestyle intervention vs. standard care in overweight/obese adults	Annual	60 years	PSA (normal, gamma, beta distributions) Monte Carlo CEACs
Green et al. (2022)	USA	Microsimulation CEA	Lifetime cost-effectiveness of screening and treating masked hypertension: usual care vs. ABPM vs. HBPM	Annual	Lifetime	One-way SA (ABPM specificity, screening frequency, pill disutility) two-way SA
Houle et al. (2012)	Canada	Cost-avoidance model PSA	Cost savings from reduced CVD events under pharmacist-led hypertension management in patients with diabetes	Annual	1 year	Monte Carlo (10,000 iterations) varied SBP reduction, effectiveness decline, pharmacist time
Hutchinson et al. (2024)	24 LMICs	Markov model	Costs and consequences of strengthening hypertension care cascade vs. status quo at three SBP cut-points	Annual	30 years	VSLY, earnings, and medication costs shown as key drivers
Johnson et al. (2018)	USA	Causal inference model	Simulated BP treatment policies applied to EHR data (2008–2016) using the parametric g formula	Not specified	2008–2016	Not reported
Ke et al. (2021)	China	Markov model	Lifetime cost-utility of integrated vs. usual care for hypertension in rural southwest China	Annual	Lifetime (65 years)	One-way deterministic SA PSA (Monte Carlo)
Keuffel et al. (2019)	USA	Decision analytic model	Hospital cost reductions from improved MAP control in septic ICU patients models AKI probability and mortality	Not applicable	Not applicable	One-way and PSA
Knight et al. (2017)	New Zealand	Monte Carlo sampling multiple imputation	Synthetic national population to evaluate CVD medication treatment policy options	Not applicable	Not applicable	Planned for future simulations
Li et al. (2021)	China	Microsimulation (Markov, bootstrapped)	Lifetime cost-effectiveness of intensive SBP treatment (<120 mmHg) vs. standard treatment in SPRINT-eligible adults	Annual	Lifetime	Deterministic SA PSA (1,000 runs)
Li et al. (2023)	USA	Microsimulation	Long-term impact of home BP monitoring vs. usual care on MI, stroke, and healthcare costs	Annual	20 years	Varied discount rates
Liao et al. (2023)	USA, UK, China	Markov model	Cost-effectiveness of intensive vs. standard BP management in patients aged 60–80 (China, US, UK)	Annual	Lifetime	One-way SA PSA (Monte Carlo, 1,000 runs)

Author (Year)	Country	Model Type	Scope	Time Steps	Time Horizon	Sensitivity Analysis
Lich et al. (2014)	USA	System Dynamics model	15 stroke prevention scenarios in US Veterans on 20-year QALYs, strokes prevented, and NNT per QALY	Annual	20 years	Multivariate PSA on intervention effect sizes and model inputs
Margolis et al. (2020)	USA	Microsimulation	Validates microsimulation-predicted CV events against 5-year HyperLink telemonitoring trial outcomes	6, 12, 18, 54 months	5 years	Adding medication costs ROI recalculated under primary and secondary endpoints
Marklund et al. (2020)	China	Comparative risk assessment (BP + potassium models)	CVD deaths, events, and DALYs prevented by replacing dietary salt with potassium-enriched substitutes (20–30% KCl), including CKD hyperkalaemia risks	Not applicable	Not applicable	PSA (Monte Carlo)
Mekonnen et al. (2013)	USA	Computer simulation (CVD and diabetes)	CVD and diabetes impact of sugar-sweetened beverage reduction in California (2013–2022)	Annual	10 years	20% and 50% SSB reduction scenarios
Murray et al. (2003)	14 world subregions	Population model	DALYs averted and costs of personal and non-personal BP/cholesterol interventions across 14 world subregions	Annual	10 years	Multivariate SA
Peñaloza-Ramos et al. (2018)	UK	Cohort Markov model (4 structures)	Structural uncertainty in the TASMIN-SR cost-effectiveness model across four alternative Markov structures	6-month cycle	Lifetime	Robust to varied costs, time horizons, and secondary event transition probabilities
Peñaloza-Ramos et al. (2016)	UK	Cohort Markov model	Long-term cost-effectiveness of BP self-management vs. usual care in high-risk 70-year-old patients	6-month cycle	Lifetime (30 years)	PSA: 10,000 Monte Carlo simulations CEACs
Stevanović et al. (2014)	Netherlands	Cohort Markov model CUA PSA	Long-term cost-effectiveness of SBP reduction via antihypertensive treatment for primary CVD prevention in mild hypertension	Annual	10-year and lifetime	PSA (10,000 iterations)
Wang et al. (2016)	China	Population-scale simulation	CVD events and costs prevented by population-wide salt restriction in China	Annual	10 years	One-way SA greater uncertainty for salt-substitute strategy
Wu et al. (2022)	China	Multi-state Markov model	Hypertension progression among Chinese bus drivers over 1- and 3-year follow-up intervals	1-year, 3-year	Not applicable	Not applicable
Xie et al. (2021)	China	Hybrid: econometric (DiD) + population simulation	EPHS population-based hypertension control policy in China evaluated using combined econometrics and simulation	Annual	10 years	One-way SA PSA
Yosefy et al. (2009)	Israel	Cross-sectional CUA	Cost-utility of expanding IBPC program to 100 clinics in Israel QALYs gained and costs saved over 10-year AMI risk horizon	Not applicable	10-year AMI risk	Not reported

Author (Year)	Country	Model Type	Scope	Time Steps	Time Horizon	Sensitivity Analysis
Yosefy et al. (2007)	Israel	Spreadsheet-based CUA deterministic	Health effects and costs of a proposed national hypertension prevention program over 20 years	Annual	20 years	SA on parameters with greatest potential influence on costs and utilities
Zhang et al. (2021)	Argentina	Cohort Markov model CEA	Cost-effectiveness of a multicomponent hypertension program in adults with uncontrolled BP in a low-income setting	6-month cycle	Lifetime	One-way SA PSA
Zhou et al. (2024)	China	Decision-analytic Markov model	Four antihypertensive initiation threshold strategies for primary CVD prevention	Annual	10 years	One-way SA (CVD incidence) PSA (hazard ratios)

Abbreviations: CEA = cost-effectiveness analysis; CUA = cost-utility analysis; CVD = cardiovascular disease; DiD = difference-in-differences; ICER = incremental cost-effectiveness ratio; LMIC = low- and middle-income country; PSA = probabilistic sensitivity analysis; SA = sensitivity analysis; SBP = systolic blood pressure; CEAC = cost-effectiveness acceptability curve; NNT = number needed to treat; QALY = quality-adjusted life year; DALY = disability-adjusted life year.

eSupplementary Table 5. Model Parameters and Results From Rapid Review (n=39 studies)

Author (Year)	Key Input Parameters	Heterogeneity / Subgroups	Key Outputs	Limitations
Almalki et al. (2020)	Transition probabilities (MI, stroke, HF, CV/non-CV death) SAE probabilities drug, visit, rehab and adverse event costs pill burden disutility	Age-specific mortality and utilities	QALYs gained ICER	Event rates from non-Saudi SPRINT population private hospital cost data patients with diabetes excluded no societal costs
Ambavane et al. (2019)	Age, SBP, stroke risk, acceptance/adherence rates, stroke and non-stroke mortality rates, birth rate	Stratified by age and SBP	Stroke cases and deaths averted cost savings breakeven intervention cost	Meditation duration not modelled only systolic hypertension individual characteristics (sex, race) not included
Basu et al. (2016)	Age, sex, urban/rural, SBP/DBP, cholesterol, diabetes, smoking BP reduction per drug class RR for MI/stroke per mmHg costs disability weights	Stratified by age, sex, location, and risk factors WHO risk equations for regional calibration	DALYs averted MI and stroke deaths averted cost-effectiveness (\$/DALY averted)	Relies on WHO risk equations focused on MI and stroke only societal costs excluded
Boersma et al. (2007)	LIFE trial population data medication costs stroke costs and incidence	Not stratified	ICER (cost per LYG)	Trial data with limited timeframe LYGs/QALYs extrapolated healthcare sector perspective only
Bryant et al. (2021)	Staff wages, reimbursements, certification, pharmacist incentive, POC testing medication adherence, adverse drug reaction probability, CVD event probability	Not stratified	QALYs ICER	Long-term outcomes uncertain societal costs excluded limited generalisability beyond LA County
Bryant et al. (2023)	RR reductions from BP lowering treatment intensification probability visit frequency wage-based costs linear adherence decline post-intervention	Age, sex, and baseline SBP ethnicity and SES not modelled	SBP reduction BP control rates CVD events cost/QALY QALYs gained	Limited racial/ethnic diversity in trials social determinants not modelled TBC workflow variation not isolated
Burke et al. (2024)	BP treatment effects (SPRINT/JNC-8) ASCVD and dementia risk equations NHANES baseline population GCP cognitive impairment threshold	Age, sex, race/ethnicity SES not modelled	ASCVD events dementia incidence cognitive performance years in optimal brain health	No J-curve imprecise dementia incidence estimates adherence not modelled
Burke et al. (2024)	Age, sex, race/ethnicity, education SBP, DBP, cholesterol, HbA1c, BMI, smoking GCP BP lowering RR age-based QALY weights	Age, sex, race/ethnicity, education, baseline risk factors racial disparities included	Stroke, MI, dementia incidence CV and non-CV mortality GCP trajectories QALYs	Linear BP effects limited Hispanic data computationally intensive ongoing updates needed
Cazarim et al. (2018)	Direct medical costs (consultations, labs, medications, emergency care) indirect costs BP control per JNC-8 costs adjusted to 2015 values	Age and cardiovascular risk profile	BP control rate cost per patient/year CER ICER INB	Missing patient data limited to Brazilian PHS context
Chay et al. (2024)	QRISK3 scores intervention and hospitalisation costs GBD disability weights transition probabilities 3% discount rate	Stratified by sex QRISK3 includes age, sex, ethnicity, SBP, diabetes, cholesterol	DALYs averted CVD events ICERs INMBs	Trial underpowered for sex-based differences constant CVD risk assumed LMIC generalisability limited
Choi et al. (2016)	NHANES sodium intake BP and treatment status MI/stroke history NSRI targets equations linking sodium→SBP→MI/stroke (Framingham)	Age, sex, race/ethnicity, income differential salt sensitivity	Hypertension prevalence MI and stroke incidence and mortality	NHANES recall bias no CEA non-BP sodium effects excluded

Author (Year)	Key Input Parameters	Heterogeneity / Subgroups	Key Outputs	Limitations
Fishman et al. (2013)	BP control rates SBP/DBP changes Social Security life expectancy data microcosted intervention components	Stratified by sex	BP control improvement life-years saved cost per life-year saved ICERs	Integrated health system only no patient-incurred costs 1-year follow-up only
Fonseca et al. (2023)	Cuff and invasive SBP age, sex Framingham risk equations (CHD, stroke, AF, HF) cholesterol, HDL, BMI, smoking, diabetes CVD event costs (2019 USD)	By SBP category and degree of discordance (0–4, 5–9, 10–14, ≥15 mmHg) age, sex, BMI, diabetes	CVD events missed due to SBP underestimation (stroke, MI, coronary death, AF, HF) costs of missed events	Angiography patients only 1-year horizon DBP not modelled missing data imputed
Galani et al. (2007)	BMI, SBP, total and HDL cholesterol transition probabilities (hypertension, diabetes, stroke, CHD) utility scores complication costs lifestyle effect sizes	Age (25, 35, 45, 55), sex, BMI category no SES or ethnicity	Life-years QALYs incremental costs ICERs	No Swiss-specific BMI-complication data smoking and some complications excluded comorbidity independence assumed
Green et al. (2022)	BP trajectories diagnostic accuracy CVD risk treatment effect adverse event probabilities screening, device, medication, and CVD event costs QoL	Age, sex, BP status no SES, ethnicity, or regional stratification	CVD events prevented serious adverse events ICERs probability of cost-effectiveness	ABPM assumed 100% sensitivity/specificity no RCTs in masked hypertension HF not modelled
Houle et al. (2012)	SBP reduction 6-month absolute risk reductions (MI, stroke, HF) inpatient event costs program and personnel costs	Based on 8 RCTs with varied patient characteristics age, sex, SES, ethnicity not explicitly modelled	Absolute risk reduction cost savings from avoided events net savings	Outpatient drug costs excluded provincial health payer perspective only likely underestimates savings
Hutchinson et al. (2024)	BP-lowering effects intervention financial costs screening, supply chain and programme costs economic value of health improvements	Not stratified	Averted stroke and MI events, deaths, DALYs averted health expenditures productivity gains	Comorbidities not modelled only stroke and MI labour productivity inputs from high-income settings
Johnson et al. (2018)	ICD-9 billing codes BP measurements NLP physician notes antihypertensive prescriptions	HR by sex and ancestry (Hispanic, Native American, African American, Asian, Caucasian)	SBP target <120 mmHg significantly reduces MACE incidence	Death excluded from MACE drug dosage changes not modelled BMI not considered unmeasured confounders possible
Ke et al. (2021)	Human/project costs per group (control, G1, G2) treatment costs per hypertension stage event costs (MI, CHF, stroke, ESRD)	Not stratified	Option 1: ICER \$3,374/QALY Option 2: dominant (ICER −\$3,681/QALY) Option 2 cost-effective at WTP \$1,599/QALY (81.3%)	SF-36 to EQ-5D mapping (UK population) sensitivity ranges set to ±20% where CIs unavailable
Keuffel et al. (2019)	P(AKI) P(death AKI or no AKI) OR of AKI and death per 1-unit MAP increase AKI costs with/without RRT sepsis hospitalisation costs	Not stratified	Cost savings: \$417,000/year (5 mmHg MAP increase) and \$600,000/year (10 mmHg) per 1,000 ICU sepsis cases	Sepsis-2 definition used MAP target of 65 mmHg may not suit patients with prior hypertension
Knight et al. (2017)	Age, sex, ethnicity, SES, smoking, CVD history, diabetes, BP/lipid-lowering medication, SBP, TC:HDL ratio, family history of CVD	Not stratified	Synthetic population of 2,451,278 individuals created and validated	Medication and family history variables not externally validated

Author (Year)	Key Input Parameters	Heterogeneity / Subgroups	Key Outputs	Limitations
Li et al. (2021)	Antihypertensive medication costs RR of CVD death, stroke, SAE, and HF from treatment SAE and post-stroke care costs	Less favourable ICERs for men (\$12,259/QALY) and SBP 130–139 mmHg (\$13,277/QALY)	>2.03M CVD deaths prevented and 3.84M life-years saved over 5 years ICER Int\$10,997/QALY (82.8% cost-effective at 1× GDP per capita)	Model relied on US SPRINT data limited SAE data BP measurement methods differed competing risks may confound estimates
Li et al. (2023)	Age, race/ethnicity, rural/urban health behaviours comorbidities (diabetes, high cholesterol) CVD history	Larger MI/stroke reductions in non-Hispanic Black and Hispanic adults, and in women	MI and stroke cases averted healthcare costs saved per person	Hypertension control as binary variable SES excluded BRFSS data self-reported intervention costs not estimated
Liao et al. (2023)	CHD, stroke, HF, AF, CV and non-CV death event rates	Lower ICERs for men ICERs varied by age group and country	ICERs: \$12,362/QALY (China), \$25,417/QALY (US), \$7,004/QALY (UK). Cost-effective in all three settings.	Han participants only repeated CVD risks not modelled kidney outcomes excluded
Lich et al. (2014)	Time delays, constants, rates, and time series inputs	Not stratified	Hypertension control: 35.517 QALYs (largest). Targeted CVD prevention: 27,856. Most efficient NNT: thrombolytics (3.1), rehabilitation (3.9)	Costs not directly included (NNT used as efficiency proxy) prevention benefits may be underestimated
Margolis et al. (2020)	BP measurements CV event incidence costs observed SBP difference between TI and UC groups at 12 and 54 months	Not stratified	CV events: 4.4% (TI) vs 8.6% (UC), OR=0.49. Net savings: \$1,900/patient over 5 years (ROI 126%)	Small single-site trial CV event reduction not statistically significant high intervention costs
Marklund et al. (2020)	Pre-intervention SBP SBP effect on CVD/CKD risk CVD/CKD burden CKD prevalence by stage CVD mortality HR in CKD potassium intake and serum effect	Not stratified	~461,000 CVD deaths prevented/year (11%) ~7.9M DALYs saved ~11,000 additional CKD hyperkalaemia deaths net: ~450,000 fewer CVD deaths	BP estimates from stroke/hypertensive patients only CEA not conducted no time component or recurring events modelled
Mekonnen et al. (2013)	SSB serving size caloric compensation RR of diabetes per ≥1 SSB/day BMI effects on SBP, cholesterol, diabetes price elasticity	Greater benefits for African Americans, Mexican Americans, and low-income populations	10–20% SSB reduction: 12,000–23,000 fewer incident diabetes cases (1.8–3.4%) over 10 years 50% reduction: 53,000 fewer cases (8.0%)	SSB-health links from observational data self-reported SSB intake societal costs and lipid effects excluded
Murray et al. (2003)	Mass media education salt reduction cholesterol and BP treatment (2 thresholds each) absolute CVD risk-based treatment (4 thresholds)	Not stratified	Non-personal interventions: >21 million DALYs averted/year globally. Combination treatment (>35% 10-year risk): 63 million additional DALYs averted/year.	Mass-media effect estimates from high-income settings salt reduction benefits debated long-term compliance uncertain
Peñaloza-Ramos et al. (2018)	CVD, HF, MI, stroke, TIA, UA health states TASMIN-SR trial transition probabilities	Not stratified	Self-management dominant across all four model structures. EVPI substantially reduced for Model 1.	Limited to cohort Markov models no formal methods to assess model plausibility
Peñaloza-Ramos et al. (2016)	SBP reduction at 6 and 12 months annual transition probabilities for stroke, MI, UA (DM, CKD, prior stroke, CHD subgroups)	Not stratified	Self-management dominant for men and women. 99% cost-effective at £20,000/QALY 97% at £10,000/QALY.	Trial participants may differ from non-participants subsequent CVD events not modelled

Author (Year)	Key Input Parameters	Heterogeneity / Subgroups	Key Outputs	Limitations
Stevanović et al. (2014)	CVD incidence (stroke, MI, HF, unstable angina) SBP reduction from HCT 25mg adherence drug costs GP visits discount rates life expectancy	ICERs by age and sex: men €6,032–58,217/LYG women €12,345–361,064/LYG at 10-year horizon	ICERs at 10-year horizon (men and women by age and SBP). Lifetime estimates more favourable for both sexes.	SCORE predicts CVD mortality only instant SBP reduction assumed adverse drug effects excluded societal costs excluded
Wang et al. (2016)	Population size CVD risk factor distributions CHD and stroke incidence, case fatality, and mortality stratified by age group, sex, and region	Not stratified	Annual CVD events prevented CVD incidence and mortality reductions QALYs gained treatment costs saved	BP-lowering effects may be overestimated in normotensives intervention investment costs excluded
Wu et al. (2022)	Height, weight, BP, blood glucose, TG, cholesterol, creatinine, uric acid age, sex, smoking history	Increasing age, BMI, cholesterol, TG as risk factors for BP progression in males (HR: 1.04–2.53)	1-year P(normal→hypertension): 4.60% male, 2.13% female 3-year: 20.27% male, 7.98% female	Single-workplace recruitment (selection bias) hypertensive state treated as absorbing
Xie et al. (2021)	Population age hypertension prevalence GDP/capita EPHS coverage RR per 10 mmHg SBP reduction incidence/mortality rates per-person costs utility values	Not stratified	97,100 fewer MI cases and 215,600 fewer stroke cases over 10 years \$2,131/QALY saved	Assumes simulation only needed after favourable econometric results delayed benefits may be overlooked
Yosefy et al. (2009)	Age, sex, BMI, SBP, DBP, glucose, triglycerides, LDL-c, HDL-c	10-year AMI risk: 13.8% (males) and 3.9% (females). Post-treatment reduction: 11.2% in males, 14.7% in females.	Health service cost savings: \$977,993 602 QALYs gained	Non-randomised physician selection no central laboratory monetary estimates may vary
Yosefy et al. (2007)	Unit treatment costs epidemiological, treatment efficacy, and demographic inputs DALYs lost from hypertension-related morbidity and mortality	Not stratified	Annually averted: 8,749 AMIs, 8,134 PVDs, 8,074 CVAs, 2,316 ESRD. Deaths prevented: 2,242. Life-years saved: 5,117. DALYs saved: 18,726.	AMI/UAP deaths counted only within 1 year post-hospitalisation in-hospital stroke deaths only ESRD from dialysis patients only
Zhang et al. (2021)	CVD event/follow-up costs health service costs by group CVD event probability by risk P(fatal event) QoL weights QALY difference RR reduction discount rate	Not stratified	HCPIA: 8.42 QALYs, INT \$3,096 ICER INT \$4,907/QALY. Usual care: 8.29 QALYs, INT \$2,473.	Limited external validity no long-term Argentine epidemiological data lifetime benefit extension may be overly optimistic
Zhou et al. (2024)	CVD incidence rates hazard ratios of antihypertensive interventions baseline CVD risk by BP threshold strategy	Analyses by initiation threshold (BP ≥130, ≥140, ≥150, ≥160 mmHg)	Strategy 2 vs 1: 666 more CVD events prevented, NNT +10. Strategy 3 vs 1: 388 more events prevented, NNT –6. Strategy 4 vs 1: 193 fewer events prevented, NNT –18.	Chinese population only adherence and adverse effects not modelled 10-year horizon may underestimate lifetime benefits

Abbreviations: AF = atrial fibrillation; AKI = acute kidney injury; BP = blood pressure; CHD = coronary heart disease; CVD = cardiovascular disease; DALY = disability-adjusted life year; ESRD = end-stage renal disease; GBD = Global Burden of Disease; HF = heart failure; HBPM = home blood pressure monitoring; HR = hazard ratio; ICER = incremental cost-effectiveness ratio; INB = incremental net benefit; INM = incremental net monetary benefit; LMIC = low- and middle-income country; MACE = major adverse cardiovascular event; MAP = mean arterial pressure; MI = myocardial infarction; NNT = number needed to treat; NSRI = National Salt Reduction Initiative; PSA = probabilistic sensitivity analysis; QALY = quality-adjusted life year; RR = relative risk; RRT = renal replacement therapy; SA = sensitivity analysis; SBP = systolic blood pressure; SES = socioeconomic status; SSB = sugar-sweetened beverage; TBC = team-based care; WTP = willingness-to-pay threshold.

eSupplementary Table 6. SBP distribution by 5-year age groups and sex based on NDHS 2022.

Age Category	Sex	Mean (mmHg)	Std Dev
0	female	103.9	12.3
5	female	103.9	12.3
10	female	103.9	12.3
15	female	103.9	12.3
17	female	103.9	12.3
20	female	104.8	12.6
25	female	106.3	14.7
30	female	109	17.4
35	female	112.3	16.9
40	female	116.3	25.6
45	female	120.9	22.8
50	female	124.7	24.8
55	female	126.9	28.6
60	female	125.8	25.4
65	female	129.7	26.9
70	female	130.5	25.1
75	female	134.9	26.6
80	female	144.5	30.1
85	female	144.5	30.1
0	male	108.9	12.6
5	male	108.9	12.6
10	male	108.9	12.6
15	male	108.9	12.6
17	male	108.9	12.6
20	male	112.5	15.5
25	male	115	12.4
30	male	117.3	16.6
35	male	120.6	21.3
40	male	120.1	22.2
45	male	126.9	24.4
50	male	127.4	27.6
55	male	124.3	24.5
60	male	124.7	23.4
65	male	129.5	24.9
70	male	132.4	23
75	male	128.1	27.4
80	male	133.2	24.7
85	male	133.2	24.7

eSupplementary Table 7. Business-as-usual input parameter and epidemiological parameters and their sources for the base year (i.e. 2023) and projections.

Parameter	Data Source	Model assumptions
Intervention		
Distribution of individuals by age and sex	NSO (2025)	Data on mean predicted SBP by age and sex were used to specify the SBP distribution for baseline population in 2023, using data from Nepal Demographic Health Survey. We used a baseline population of 29.1 million based on Nepal Statistics Office (NSO) stratified by sex for baseline in 2023.
Time exposed (Y)	Assumed	We assumed shifting SBP distribution impacts individual's total exposure history. This time exposed was then used with the intervention effect size.
Actual intervention = Δ SBP (mmHg)	(Author specified)	For those with high SBP (≥ 140 mmHg), the actual intervention was defined as shifting mean SBP to < 140 mmHg.
Time lags	Assumed	Because there are delays in how diseases develop over time, eradicating high SBP (≥ 140 mmHg) won't immediately affect the occurrence of diseases. Though we don't have precise information on the exact duration of these delays, prior evidence suggests that high SBP (≥ 140 mmHg) associated changes in chronic disease (e.g. Cancer, CVD) incidence was seen as early as three years post intervention). (24, 40) To address this uncertainty, we incorporated a time delay of 0 to 3 years for CHD (In normal $\pm 20\%$), and stroke.
BAU – epidemiological parameters		
All-cause mortality rates	GBD 2023	Data on all-cause mortality rates by sex and age group for 2023 were obtained from the Global Burden of disease results tool.(41) Data on SES rate ratio differences in all-cause mortality is not available, therefore we didn't consider stratified analysis by SES (see Methods text below).
All-cause morbidity rates	GBD 2023	Data on years lived with disability (YLD) for cardiovascular disease (CVD) were obtained from the Global Burden of Disease (GBD) 2023 study, stratified by age group and sex. No temporal trends were modelled, and 2023 YLD rates were assumed constant over the simulation period. As SES-specific morbidity rates were not available for Nepal, we didn't consider stratified analysis by SES (see Methods text below).
Disease specific incidence, prevalence, and case fatality rates	GBD 2023	Sex by age: We applied Nepal disease-specific estimates from the GBD 2023 study (24, 40) to the population of Nepal in the model. The disease-specific incidence rates, prevalence and mortality rates, and case fatality rates (mortality rate divided by prevalence), by sex and age, for all diseases included in our model were obtained from the GBD data. Disease-specific incidence, prevalence, mortality, and case-fatality rates were estimated in a coherent framework consistent with national Global Burden of Disease (GBD) 2023 estimates (24, 40). Uncertainty: SD was calculated by dividing the log of length of uncertainty interval (upper limit- lower limit) for disease specific incidence, prevalence, and case fatality rates by 3.92. For rate ratios by sex, we applied

Parameter	Data Source	Model assumptions
		uncertainty +/- 5% SD (log normal distribution for incidence, remission, fatality and beta for prevalence), correlations 1.0 between sexes, and age groups for all diseases. Annual Percentage Change (APC): the APC was estimated by fitting a least squares regression line to the natural logarithm of incidence rates and case fatality rates from 1990 to 2023 GBD data and included as inputs to the BP-MSLT out to year 2035, then rates held constant. (We assume the APC was constant across sexes.)
Disease specific morbidity	GBD 2023	The sex and age specific disability rates were calculated as disease's YLD obtained from GBD (41) divided by the prevalent cases. The same disability rate was assumed by sex (i.e., those with disease are assumed to have same severity distribution across sexes). <i>Uncertainty: +/- 10% SD. We used lognormal (incidence, remission) and beta (prevalence) distribution where appropriate.</i>
SBP-related disease	GBD 2023 (24)	Five SBP related diseases were included (ischaemic heart disease, stroke, hypertensive heart disease, chronic kidney disease, atrial fibrillation).
RRs for SBP-related disease	Global Burden of Disease (24, 40)	RRs for SBP related diseases were obtained from GBD 2023. <i>Uncertainty: $SE(\log RR) = (\log(Upper\ CI) - \log(Lower\ CI)) / 3.92$. We modelled this as lognormal distribution.</i>

eSupplementary Table 8. Future HALYs gained from achieving 100% hypertension control (SBP <120 mmHg) among individuals with diagnosed hypertension compared to BAU in Nepal.

Health adjusted life years	Undiscounted (95% UI) ^a		Discounted (95% UI) ^a at 3% per annum	
	2023-2043	Lifetime	2023-2043	Lifetime
Achieving SBP <130 mmHg				
Overall				
Total HALYs (millions)	1.9 (1.3 to 2.7)	38.5 (26.6 to 54.7)	1.3 (0.9 to 1.8)	9.4 (6.6 to 13.0)
Total HALYs per 1000 people in 2023	69.5 (50.0 to 97.7)	1386.1 (975.5 to 1970.3)	45.8 (32.9 to 64.5)	337.5 (239.6 to 469.5)
By sex				
Females, per 1000 people in 2023	57.0 (39.9 to 79.8)	1201.5 (829.5 to 1685.8)	37.6 (26.3 to 52.8)	282.7 (196.1 to 394.1)
Males, per 1000 people in 2023	83.0 (60.6 to 116.0)	1581.3 (1108.2 to 2204.7)	54.6 (39.9 to 76.5)	394.5 (283.4 to 540.8)
Achieving SBP <120 mmHg				
Overall				
Total HALYs (millions)	2.4 (1.8 to 3.4)	50.1 (35.3 to 70.2)	1.6 (1.2 to 2.2)	12.1 (8.7 to 16.7)
Total HALYs per 1000 people in 2023	89.4 (65.4 to 120.7)	1813.2 (1277.8 to 2527.3)	59.1 (43.1 to 79.8)	440.8 (315.8 to 602.5)
By sex				
Females, per 1000 people in 2023	73.3 (52.6 to 100.3)	1556.5 (1091.2 to 2171.8)	48.3 (34.6 to 66.2)	367.2 (261.7 to 508.1)
Males, per 1000 people in 2023	106.0 (79.0 to 144.3)	2071.4 (1484.8 to 2883.9)	69.8 (52.0 to 94.9)	512.4 (374.4 to 700.7)

^apoint estimates with 95% UI in brackets.